

Evaluation Framework for Multimodal Language Models Applied to Geospatial Open Data in Agriculture

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Abstract

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Keywords: keyword 1, keyword 2, keyword 3, keyword 4

1. Introduction

The rapid development of multimodal Large Language Models (LLMs) has unlocked a huge span of unprecedented applications across diverse scientific and industrial domains. Among these, smart agriculture and precision farming stand out as areas of great potential. The recent emergence of geofoundation models (Jakubik et al. (2023); Corley et al. (2026)) and Earth embeddings (Stewart et al. (2026)), coupled with the swift evolution of generative language models, holds immense promise for Earth observation and agricultural management. This progress has been significantly accelerated by open-source initiatives, which have democratized access to foundational models and driven community-led breakthroughs in applied Artificial Intelligence.

In recent years, a growing body of literature has explored the integration of generative AI into agricultural workflows. Many of these initiatives have materialized as conversational assistants tailored for Earth observation and agronomy (Yuan et al. (2025); Kuckreja et al. (2023)). These studies frequently leverage Retrieval-Augmented Generation (RAG), geo-embeddings, or hybrid approaches combining language modeling with classical computer vision tasks, such as the semantic segmentation of satellite imagery and orthophotos. However, despite these technical advances, existing approaches remain largely fragmented. While the outputs generated by these state-of-the-art tools are increasingly accurate and explore new frontiers in data accessibility, they fundamentally remain generic, failing to pro-

vide the highly localized, actionable insights required for precise, parcel-level terrain management.

To bridge this gap, we recently proposed a novel multimodal LLM framework capable of generating highly specific, localized responses regarding designated land regions for agricultural applications (Cañada et al. (2025)). This system achieves its specificity through the retrieval, processing, and fusion of diverse, multi-domain open data sources, effectively functioning as a context-amplified tool for land management.

Following the technical validation of this architecture, a critical challenge arises: the robust and methodical evaluation of its generated outputs. Since the inception of generative AI, evaluation has remained a primary and open line of research, inherent to any application relying on non-deterministic inference. Unlike traditional machine learning paradigms, where evaluation metrics rely on a finite set of ground-truth labels, the space of correct responses in generative modeling is virtually infinite—a single question can be correctly answered in multiple, equally valid ways depending on the reasoning approach. Numerous evaluation frameworks have been proposed to navigate this complexity. For instance, the LLM-as-a-judge paradigm (Zheng et al. (2023))—which we initially employed in our previous work—is widely adopted due to its simplicity and low operational cost. Nevertheless, this technique is highly vulnerable to technical hallucinations, suffers from self-reference bias, and tends to lose consistency over complex reasoning chains. Similarly, frameworks like RAGAS (Es et al. (2025)) have been developed specifically to evaluate RAG pipelines, and substantial

research has been devoted to optimizing automated evaluation metrics.

Despite this rich landscape of evaluation tools, none of the existing frameworks align with the complex requirements of our context-amplified architecture. Specifically, evaluating this system demands more than merely assessing the factual correctness of the final output; it requires interrogating how the model utilized the available external context. A rigorous evaluation must distinguish between a response that is "good but generic" and one that is "specific and context-aware." Furthermore, it must discern whether a high-quality response stems merely from the model's robust foundational pre-training, or if it is the result of successful reasoning over the dynamically retrieved multi-domain data.

To address these unmet needs, this paper proposes a novel, reproducible evaluation methodology tailored for context-amplified multimodal LLMs in agricultural data applications. Our approach introduces a multi-stage, human-in-the-loop (hybrid human-machine) framework designed to mitigate automated biases. The most challenging and innovative aspect of this framework lies in quantifying the marginal contributions of individual architectural modules. The methodology allows to understand not only the final quality of a generated response, but exactly how and to what extent each specific module influenced that quality. To achieve this, we design a comprehensive ablation study based on a factorial design, systematically evaluating different configurations by isolating, adding, and removing system modules. The evaluation results are subsequently analyzed through collaborative game theory, providing a rigorous mathematical framework to measure the precise impact and value added by each component.

The entirety of this methodology and its associated tools are open source. Furthermore, to ensure complete reproducibility and to provide a valuable asset for the research community, we have published our complete evaluation dataset on Zenodo as supplementary material. This open dataset comprises 143 detailed evaluations, containing the specific agricultural parcels, the exact external context retrieved, and the corresponding responses generated by the model.

The remainder of this article is structured as follows: Section 2 details the architecture of the conversational system, provides a functional overview, and extensively outlines the evaluation methodology, including parcel selection, human-in-the-loop judge guidelines, evaluation stages, utilized metrics, and the design of the ablation study. Section 3 presents and discusses the results derived from the evaluation and the game theory analysis. Finally, Section 4 summarizes the main conclusions drawn from this work.

2. Materials and methods

2.1. System Architecture and Overview

The proposed system integrates heterogeneous geospatial, remote sensing, and agronomic domain knowledge into a unified data processing pipeline to ground the reasoning of a Large Language Model (LLM). As illustrated in Fig. 1, the architecture is structured into two primary layers: the *User Interface*

and the *Backend Processing* pipeline. The core objective is to enrich the user's inquiry regarding a specific agricultural plot with multi-modal and biophysical context, enabling precise, evidence-based agronomic recommendations. The execution pipeline proceeds through five coordinated stages: (I) Plot Info Retrieval: The user provides a target plot identifier along with a natural language query. The system queries the geographic information system registry (SIGPAC) to retrieve plot-level attributes—including slope, elevation, surface area, perimeter, and registered land use—along with its georeferenced boundary geometry (encoded as GeoJSON). Hereafter, this piece of context is regarded as Plot Data. (II) Context Retrieval: Exploiting the extracted plot geometry, this module orchestrates two parallel multi-modal processing workflows: (II-a) Visual terrain analysis: High-resolution aerial imagery (orthophoto) covering the delineated plot is retrieved and processed by a Multimodal Large Language Model (MLLM) conditioned on the geometric and cadastral metadata, yielding a structured natural-language description of terrain topology, visible infrastructure, and surface patterns. (II-b) Biophysical state extraction: Multi-spectral imagery from the Sentinel-2 constellation (Copernicus, European Space Agency) is acquired. Following spatio-temporal filtering and band synthesis over the plot geometry, the pipeline computes a set of spectral vegetation and water indices, including NDVI and NDWI. (III) Knowledge Retrieval: In parallel with context generation, a Retrieval-Augmented Generation (RAG) subsystem queries a document repository created by the user. (IV) Data Fusion and Context Aggregation: The diverse data modalities—cadastral metadata, MLLM-derived terrain descriptions, remote sensing spectral indices, and retrieved textual context—are aligned and harmonized within an aggregator. This module serializes multi-source representations into a structured natural-language prompt context. (V) LLM Reasoning and Answer Generation: The synthesized prompt is fed to the central reasoning LLM, which integrates the holistic parcel profile to formulate the final recommendations delivered back to the user. For further details on the inference framework and technical hardware and software components, see Cañada et al. (2025).

2.2. Evaluation Framework

The architectural characteristics of the proposed system dictate the requirements for its evaluation methodology. Crucially, this framework is not conceived merely as an *ad-hoc* assessment for this specific tool, but rather as a reproducible and adaptable evaluation standard for any modular conversational agent relying on augmented external context. The primary challenge distinguishing this methodology from standard LLM benchmarks is the necessity to evaluate the actual utilization of geospatial context during response generation, going beyond mere response quality. An output that is factually correct but generic—or generated solely from the model's parametric memory—cannot be considered equivalent to a highly personalized recommendation grounded in the specific biophysical traits of a given terrain. Consequently, a comprehensive methodology combining multiple experimental phases, modular configurations, and tailored metrics has been designed.

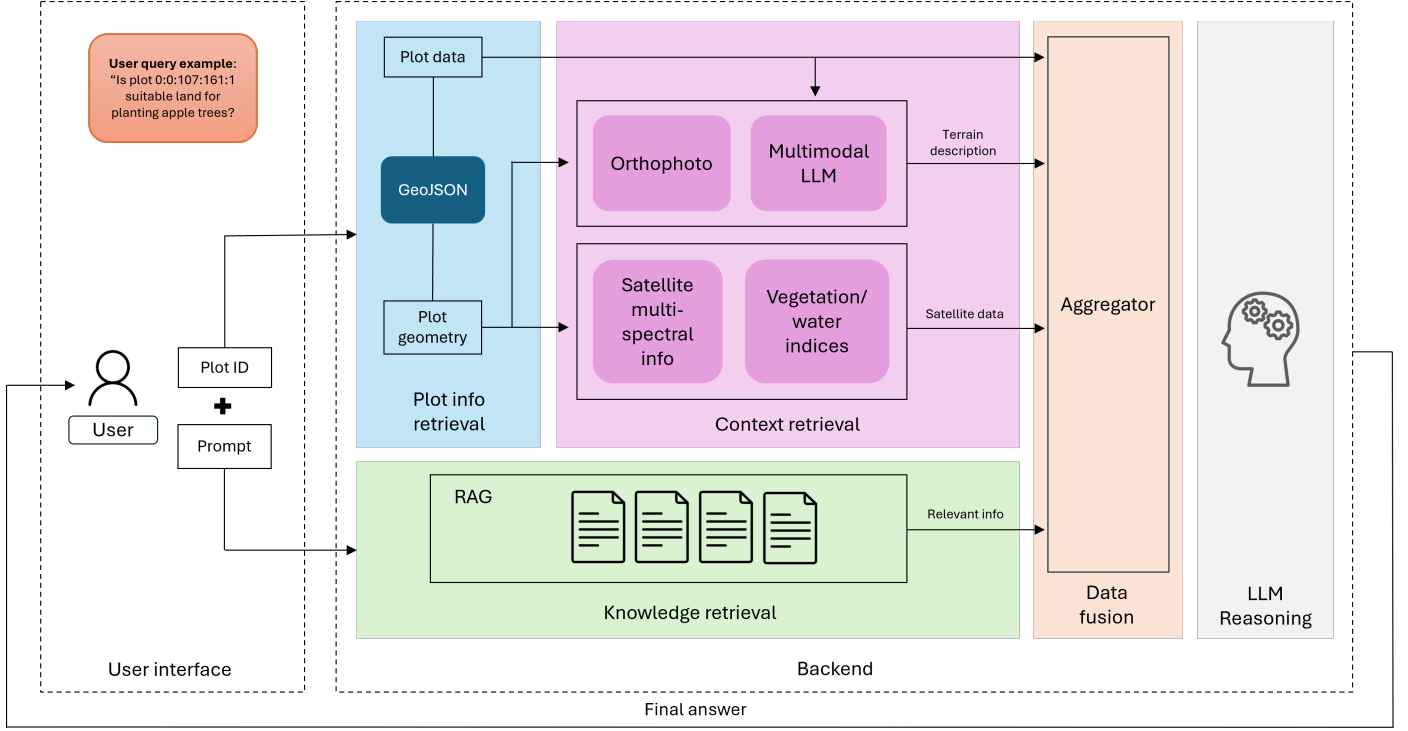


Figure 1: System architecture.

2.2.1. Parcel Selection and Heterogeneity

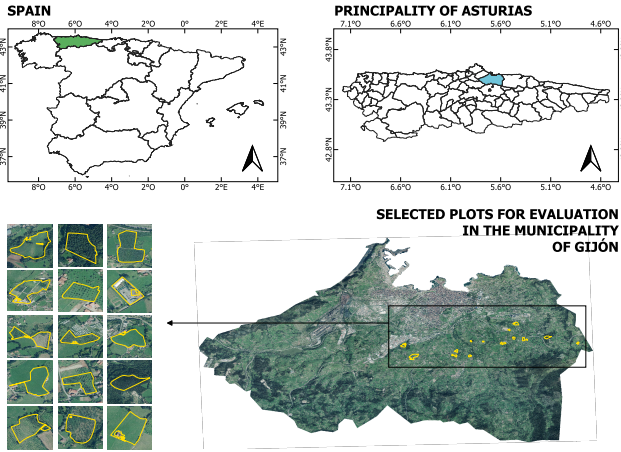


Figure 2: The area of study in Gijón, Asturias. Location of the 15 selected parcels.

The study was carried out over a total of 15 target parcels, covering four distinct Land Use (LU) types: Fruit trees (FR), Pasture (PA), Unproductive (UN) and Forest (FO). The distribution of samples within Land Uses, as well as the volume of evaluation queries per parcel, is weighted according to the operational utility of the tool for each functional land use. 7 out of the 15 parcels belong to Fruit trees category, 4 are Pastures, and the remainder Unproductive and Forest land uses get 2 parcels each. Both the number of parcels per LU and the number of queries per parcel, are detailed in Table 2. **Pendiente la parte**

de Fernando → Furthermore, the selected parcels exhibit a high degree of physical and geographical heterogeneity to stress-test the system under diverse conditions. This variance includes distinct topographies with slopes ranging from 0.5% to 21.6%, varying altitudes, highly irregular geometric boundaries, and significant diversity in vegetative health, as captured by variable NDVI ranges. The spatial distribution and morphological characteristics of these parcels are illustrated in Fig. 3. Referencia a Fig. 2.

2.2.2. Two-Stage Experimental Design

The evaluation pipeline is structured into two main sequential phases to optimize operational resources while maintaining statistical robustness: an overall performance assessment and an ablation study. The first phase, named *Broad Performance Assessment*, evaluates the system's holistic capability across diverse scenarios, using the full architecture. This stage does not represent a novel contribution, as it is similar to existing evaluation methods; rather, it provides an overall view of the quality of the responses in general terms, which is necessary before moving on to a more specific evaluation of the architecture modules. Table 2 contains the number of evaluation queries, resulting in roughly 50% for each stage.

The second phase consists of an *Ablation Study* intended to quantify the marginal contribution of each architectural component (Orthophoto, Satellite, and RAG modules, previously introduced in Fig. 1). The concept of ablation study, originally rooted in medicine and biology to describe the surgical removal of tissue to determine its physiological function, has been adapted to the field of artificial intelligence and proved as

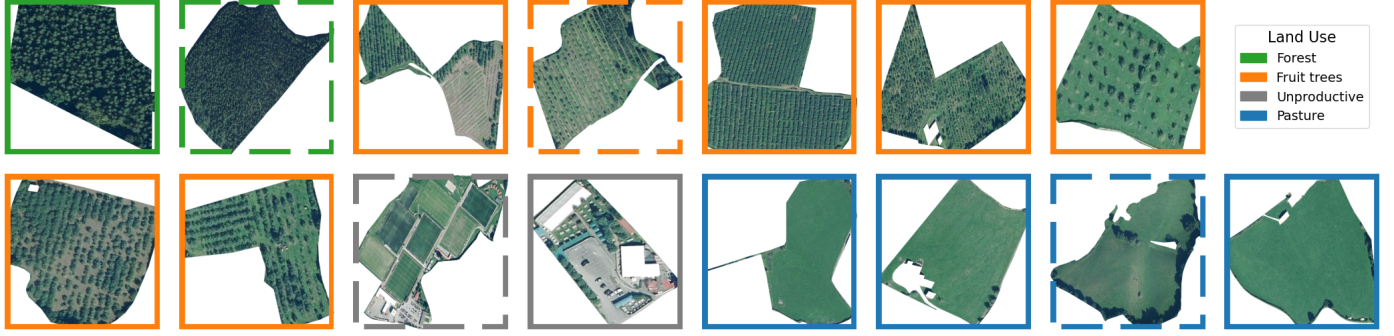


Figure 3: Selection of parcels for tool evaluation, grouped by Land Use. The number of parcels per LU depends on their functional utility in agriculture. Dashed borders correspond to parcels used in ablation study.

a valid paradigm to uncover individual contributions by isolating components. In Meyes et al. (2020), they use the concept of ablation for finding redundancies in the representations of Artificial Neural Networks. In this work, we extend this analytical framework to evaluate our proposed modular methodology by progressively isolating and bypassing individual modules. For this purpose, a full factorial design of experiments was executed to set up the configurations, which are combinations of the modules, summarized in Fig. 4. By systematically enabling and disabling these three external context modules, $2^3 = 8$ configurations were generated, plus a Zero-Shot baseline, where the LLM reasons without any external parcel context, for a total of 9 configurations, as detailed in Table 1. For this phase, a representative subset of one parcel per LU category was selected, and 2 specific queries were executed across all 9 configurations, yielding 72 additional evaluations (2).

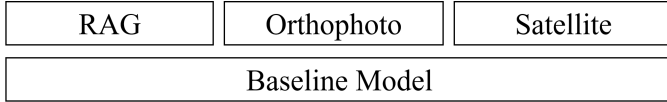


Figure 4: Ablation modules.

Ultimately, the combined phases produced a comprehensive evaluation dataset of 143. The full dataset containing queries, retrieved contexts, and generated responses is publicly available (see Appendix A). [Ref a Zenodo](#)

2.2.3. Panel of judges

A simplified approach to response evaluation, as utilized in our previous work, relies exclusively on the LLM-as-a-judge framework. This paradigm offers distinct advantages in terms of execution speed and scalability, allowing for the rapid assessment of large datasets without the bottlenecks associated with human review. Conversely, assembling a panel of human evaluators requires substantial planning, timeframes, and financial resources. However, human evaluation remains indispensable, as it fundamentally reduces the leniency bias frequently exhibited by LLM evaluators (Gu et al. (2026)) and provides a significantly higher degree of reliability when assessing nuanced, domain-specific reasoning. To effectively navigate this trade-off between scalability and qualitative rigor, we designed

a hybrid, human-in-the-loop evaluation strategy. By merging human and machine critical judgement, this methodology not only optimizes resource allocation but also systematically mitigates the mono-dimensional biases inherent to using either approach in isolation.

Consequently, the definitive panel of judges comprises four distinct entities: two human evaluators and two LLM-based evaluators. To ensure both technical depth and practical accessibility, the human contingent is divided into two distinct roles, a domain expert (holding a PhD in Biology), regarded hereafter as "Human Expert" (H_E) that provides rigorous scientific scrutiny; and a control evaluator, regarded as "Human Control" (H_C), representing the end-user perspective in terms of clarity and usability for non-academic audience. For the LLM side, we integrated two diverse language models with comparable computational footprints into the panel: a proprietary model (LLM_G), *Gemini 3.1 Flash Lite*, and an open-source model (LLM_Q), *Qwen3.5-397B-A17B-Instruct*.

2.3. Multidimensional metrics

The quality of the generated responses is quantified using a multidimensional 5-point Likert scale encompassing five distinct criteria (see Table 3). Four of these metrics—*Correctness*, *Completeness*, *Relevance*, and *Utility*—evaluate the intrinsic quality of the response and can be assessed without access to the retrieved context. Conversely, the *Grounding* metric strictly requires contextual reference; it is specifically designed to measure the degree to which the formulated response is faithfully derived from the injected multi-domain data rather than the model’s pre-existing generalized knowledge. In doing so, it serves as a critical diagnostic tool to identify and penalize *internal knowledge over-reliance*, a phenomenon where an LLM disregards or underutilizes dynamically retrieved evidence, defaulting instead to its internalized knowledge. This distinction carries profound practical implications for model selection and hardware feasibility in precision agriculture. The computational overhead of an LLM is directly related to the depth and scope of its internal knowledge base. Consequently, executing large foundation models locally on edge devices remains computationally prohibitive. If empirical evaluation reveals that domain-specific accuracy is primarily governed by robust con-

Table 1: Configurations construction using factorial design over architectural modules. PD means Plot Data. Active tool is always enabled because it is responsible for identifying parcel IDs as so.

Modules	RAG	ORTHO	SAT	PD	Active Tool	Code
Baseline	0	0	0	1	1	0
B + SAT	0	0	1	1	1	1
B + ORTHO	0	1	0	1	1	2
B + ORTHO + SAT	0	1	1	1	1	3
B + RAG	1	0	0	1	1	4
B + RAG + SAT	1	0	1	1	1	5
B + RAG + ORTHO	1	1	0	1	1	6
Complete	1	1	1	1	1	7
ZERO SHOT	0	0	0	0	1	ZS

Table 2: Distribution of parcels, queries, and configurations across methodology stages for each land use category.

Land Use	Broad			Ablation				Total
	No. parcels	Queries/parcel	No. queries	No. parcels	Queries/parcel	No. configs	No. queries	
Fruit trees (FR)	7	5	35	1	2	9	18	–
Pasture (PA)	4	5	20	1	2	9	18	–
Forest (FO)	2	4	8	1	2	9	18	–
Unproductive (UN)	2	4	8	1	2	9	18	–
Total			71				72	143

textual grounding rather than massive internal knowledge, high-performance agricultural reasoning can be achieved using significantly smaller, quantized models.

Appendix B contains the evaluation rubric given to the judges for all 5 matrices.

Table 3: Multidimensional evaluation metrics framework. Note: The *Grounding* metric measures the degree to which responses are derived from the obtained context, rather than generic knowledge, making it the only metric requiring context for evaluation.

Metric	Type
Correctness	Without Context
Completeness	Without Context
Relevance	Without Context
Utility	Without Context
Grounding	With Context

2.4. Mathematical analysis of the results

This section describes the mathematical quantification of the ablation study results. Evaluating queries across diverse architectural configurations yields empirical data on marginal contributions; however, there is yet a need to quantify the intrinsic value added by each of the three modules to the final response quality. To provide a rigorous quantification, we model the multi-modular architecture within the mathematical framework of cooperative game theory. In this paradigm, the system’s structural modules act as players collaborating to produce a unified outcome, the system’s response. By applying Shapley values (Shapley (1953)), a foundational concept from cooperative

game theory that has been widely adapted for feature attribution in explainable AI with the SHAP values (Lundberg and Lee (2017)), we can fairly and precisely distribute the total system performance among its constituent components based on their true marginal contributions across all possible combinations. Formally, let N be the set of all players. In our architecture, the players correspond to the three primary modules: $N = \{M_S, M_O, M_R\}$. Let S denote any subset, or *coalition*, of these modules. The general Shapley value $\phi_i(v)$ for a module i given a performance value function v is computed as:

$$\phi_i(v) = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!(|N| - |S| - 1)!}{|N|!} (v(S \cup \{i\}) - v(S)) \quad (1)$$

By instantiating this general formula for our specific three-player game, we isolate the exact contribution of each module across all possible permutations:

$$\begin{aligned} \phi_S(v) = & \frac{1}{3} [v(\{M_S\}) - v(\emptyset)] \\ & + \frac{1}{6} [v(\{M_S, M_O\}) - v(\{M_O\})] \\ & + \frac{1}{6} [v(\{M_S, M_R\}) - v(\{M_R\})] \\ & + \frac{1}{3} [v(\{M_S, M_O, M_R\}) - v(\{M_O, M_R\})] \end{aligned} \quad (2)$$

$$\begin{aligned}
\phi_O(v) = & \frac{1}{3}[v(\{M_O\}) - v(\emptyset)] \\
& + \frac{1}{6}[v(\{M_S, M_O\}) - v(\{M_S\})] \\
& + \frac{1}{6}[v(\{M_O, M_R\}) - v(\{M_R\})] \\
& + \frac{1}{3}[v(\{M_S, M_O, M_R\}) - v(\{M_S, M_R\})]
\end{aligned} \tag{3}$$

$$\begin{aligned}
\phi_R(v) = & \frac{1}{3}[v(\{M_R\}) - v(\emptyset)] \\
& + \frac{1}{6}[v(\{M_S, M_R\}) - v(\{M_S\})] \\
& + \frac{1}{6}[v(\{M_O, M_R\}) - v(\{M_O\})] \\
& + \frac{1}{3}[v(\{M_S, M_O, M_R\}) - v(\{M_S, M_O\})]
\end{aligned} \tag{4}$$

By substituting the conceptual coalitions with the empirical performance scores obtained from their corresponding configurations (Table 1), we rewrite the equation in terms of configuration performances:

$$\begin{aligned}
\phi_S(v) = & \frac{1}{3}[v(C_1) - v(C_0)] + \frac{1}{6}[v(C_3) - v(C_2)] \\
& + \frac{1}{6}[v(C_5) - v(C_4)] + \frac{1}{3}[v(C_7) - v(C_6)]
\end{aligned} \tag{5}$$

$$\begin{aligned}
\phi_O(v) = & \frac{1}{3}[v(C_2) - v(C_0)] + \frac{1}{6}[v(C_3) - v(C_1)] \\
& + \frac{1}{6}[v(C_6) - v(C_4)] + \frac{1}{3}[v(C_7) - v(C_5)]
\end{aligned} \tag{6}$$

$$\begin{aligned}
\phi_R(v) = & \frac{1}{3}[v(C_4) - v(C_0)] + \frac{1}{6}[v(C_5) - v(C_1)] \\
& + \frac{1}{6}[v(C_6) - v(C_2)] + \frac{1}{3}[v(C_7) - v(C_3)]
\end{aligned} \tag{7}$$

Through this formulation, the outputs of the ablation configurations are transformed into distinct, quantified metrics that represent the definitive analytical weight of each architectural module. Furthermore, we exploit the inherent mathematical property of additive efficiency of Shapley values by formally defining the *Global Delta* as the sum of the three individual modular contributions. This metric encapsulates the total aggregate gain in the system’s performance. In practical terms, it quantifies the exact quality uplift achieved strictly through the integration of the external data sources, serving as measure to compare against conventional language models.

3. Results and Discussion

// TODO comentar sobre suficiente y necesario

Table 4: Average score of the 71 broad evaluation queries.

Metric	H_C	H_E	LLM_Q	LLM_G
Correctness	4.690	4.704	4.817	4.746
Completeness	4.803	4.817	4.944	4.958
Relevance	4.549	4.817	4.831	4.930
Utility	4.704	4.761	4.859	4.944
Grounding	4.606	4.901	4.296	4.549

This section presents the results drawn from the evaluation of the 4 judges over the 143 queries along the 5 metrics, divided into the broad evaluation and the ablation study. Firstly, ?? contains the average scores of all 71 broad evaluation queries **meter (+- std) en la tabla?**.

Similarly, the trend across all metrics is ascending along the x-axis (values are lower in left-side configurations), showing how, for example, system utility in the *Zero Shot* configuration is below 1/5 for most evaluators. Since this configuration corresponds to a standard LLM model without context augmentation, and with configurations progressively ordered by active modules, it can be concluded that **responses, as expected, improve as more geospatial terrain information is added**. This result consequently verifies the initial hypothesis and validates the proposed architecture.

More generally, ?? represents the average metric values from the *Broad* stage evaluations, i.e., the complete system. For all evaluators, the resulting metrics average values very close to 5/5, implying that the system has responded with a high degree of success to the evaluated queries. These preliminary results serve as evidence that the proposed system is capable of generating high-quality, complete, useful, and generally successful responses in the broad sense.

Meter gráficas solo con valores del promedio de los 4 jueces para no crear ruido visual

Referencia a zenodo con dataset anonimizado

4. Conclusions

Acknowledgements

Thanks to ...

Appendix A. Appendix A: Shapley values

Appendix B. Appendix B: Evaluation rubric

Appendix B.1. Without Context Metrics

Table B.6 outlines the metrics used to evaluate responses without relying on external context: Correctness, Completeness, Relevance, and Utility.

Appendix B.2. With Context Metrics

Table B.7 details the grounding metric, which evaluates the response based on the provided external context.

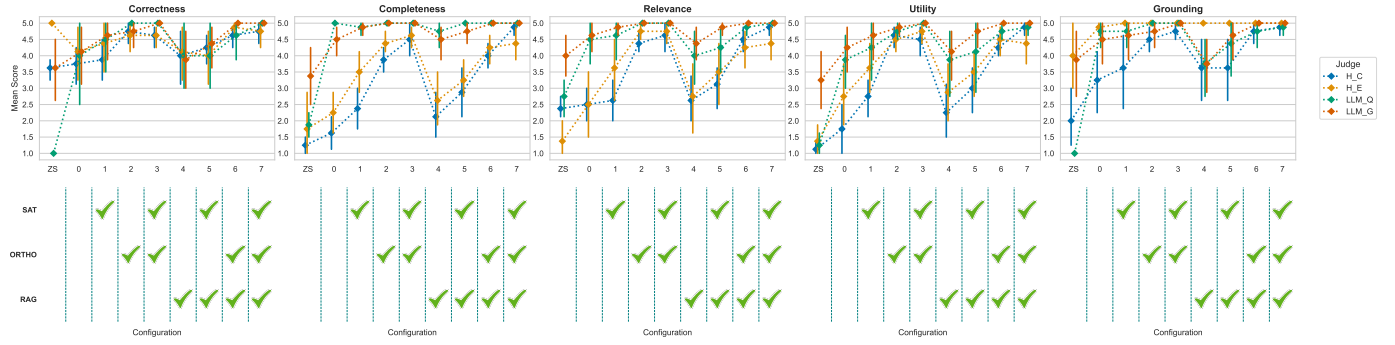


Figure 5: Preliminary evaluation results from the *Ablation Study* phase for the 4 evaluators and 5 metrics grouped by configuration. Differences can be observed based on active modules.

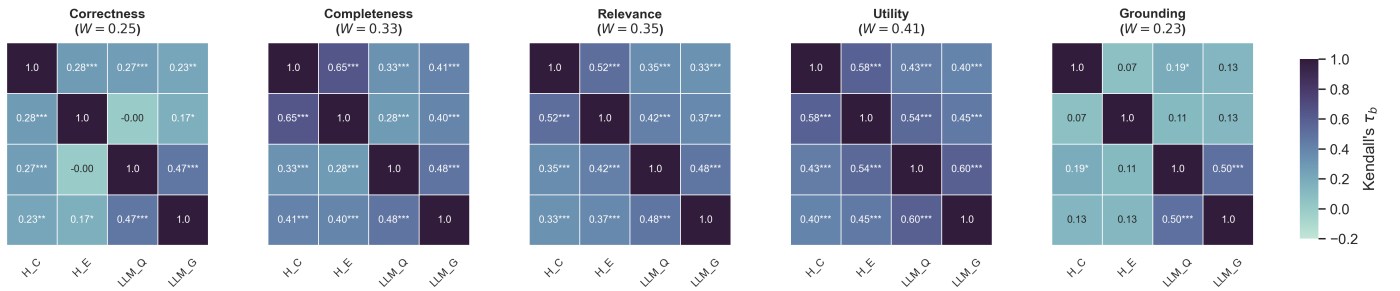


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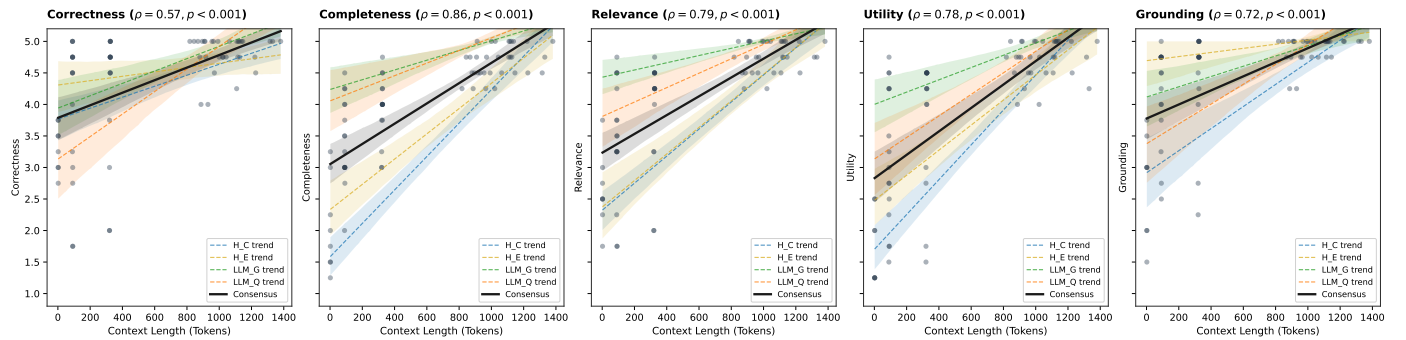


Figure 7:

References

- Cañada, J., Alonso, R., Molleda, J., Díez, F., 2025. A multimodal conversational assistant for the characterization of agricultural plots from geospatial open data, in: 2025 4th International Conference on Geographic Information and Remote Sensing Technology (GIRST), pp. 127–132. doi:10.1109/GIRST67753.2025.11382141.
- Corley, I., Lehmann, N., Robinson, C., Tseng, G., Fuller, A., Alemohammad, H., Shelhamer, E., Marcus, J., Kerner, H., 2026. No one knows the state of the art in geospatial foundation models. URL: <https://arxiv.org/abs/2605.12678>, doi:10.48550/ARXIV.2605.12678. version Number: 1.
- Es, S., James, J., Espinosa-Anke, L., Schockaert, S., 2025. Ragas: Automated Evaluation of Retrieval Augmented Generation. URL: <http://arxiv.org/abs/2309.15217>, doi:10.48550/arXiv.2309.15217. arXiv:2309.15217 [cs.CL].
- Gu, J., Jiang, X., Shi, Z., Tan, H., Zhai, X., Xu, C., Li, W., Shen, Y., Ma, S., Liu, H., Wang, S., Zhang, K., Lin, Z., Zhang, B., Ni, L., Gao, W., Wang, Y., Guo, J., 2026. A survey on LLM-as-a-judge. The Innovation 7, 101253. URL: <https://linkinghub.elsevier.com/retrieve/pii/S2666675025000000>, doi:10.1016/j.xinn.2025.101253.
- Jakubik, J., Roy, S., Phillips, C.E., Fraccaro, P., Godwin, D., Zadrozny, B., Szwarcman, D., Gomes, C., Nyirjesy, G., Edwards, B., Kimura, D., Simumba, N., Chu, L., Mukkavilli, S.K., Lambhate, D., Das, K., Bangalore, R., Oliveira, D., Muszynski, M., Ankur, K., Ramasubramanian, M., Gurung, I., Khallaghi, S., Hanxi, Li, Cecil, M., Ahmadi, M., Kordi, F., Alemohammad, H., Maskey, M.,

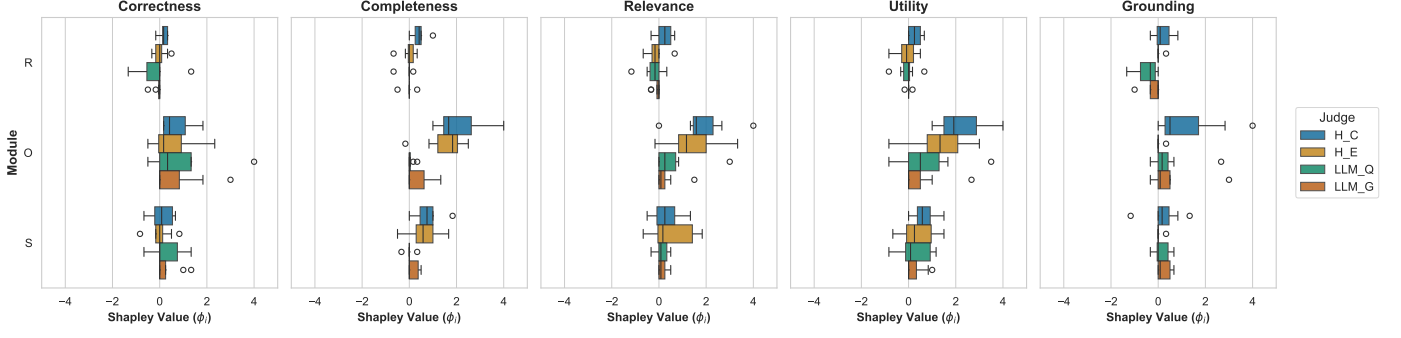


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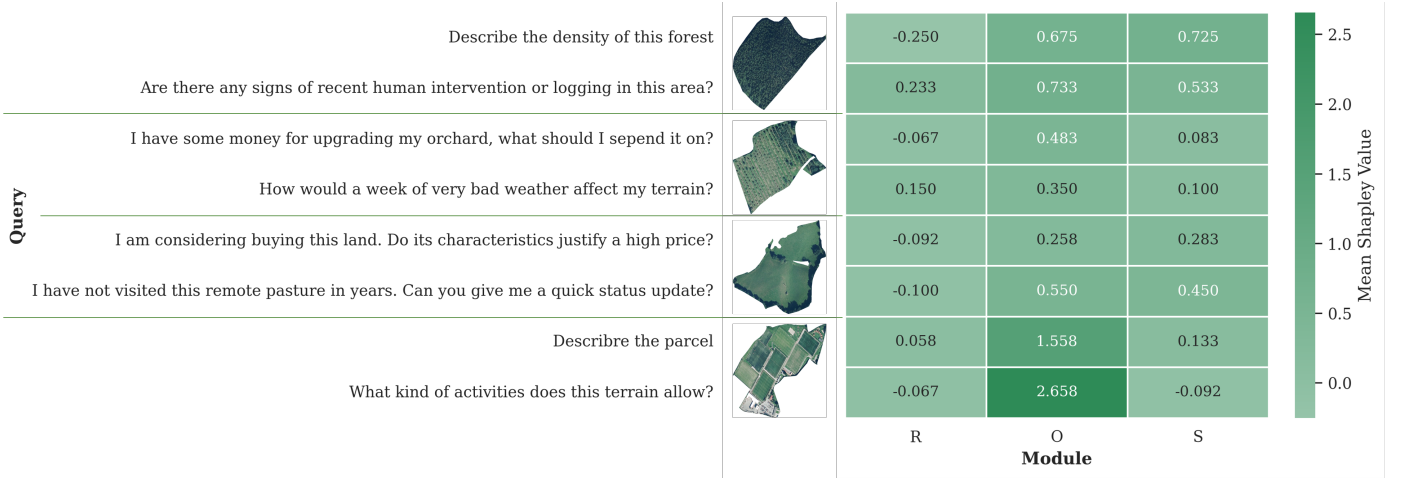


Figure 9:

- Ganti, R., Weldemariam, K., Ramachandran, R., 2023. Foundation Models for Generalist Geospatial Artificial Intelligence. URL: <http://arxiv.org/abs/2310.18660>, doi:10.48550/arXiv.2310.18660. arXiv:2310.18660 [cs].
- Kuckreja, K., Danish, M.S., Naseer, M., Das, A., Khan, S., Khan, F.S., 2023. GeoChat: Grounded Large Vision-Language Model for Remote Sensing. URL: <http://arxiv.org/abs/2311.15826>, doi:10.48550/arXiv.2311.15826. arXiv:2311.15826 [cs].
- Lundberg, S., Lee, S.I., 2017. A Unified Approach to Interpreting Model Predictions. URL: <http://arxiv.org/abs/1705.07874>, doi:10.48550/arXiv.1705.07874. arXiv:1705.07874 [cs.AI].
- Meyes, R., Lu, M., Waubert de Puiseau, C., Meisen, T., 2020. Ablation Studies to Uncover Structure of Learned Representations in Artificial Neural Networks.
- Shapley, L.S., 1953. A value for n-person games, in: Kuhn, H.W., Tucker, A.W. (Eds.), Contributions to the Theory of Games II. Princeton University Press, Princeton, NJ. volume 28 of *Annals of Mathematics Studies*, pp. 307–317.
- Stewart, A.J., Fang, H., Corley, I.A., Zhu, X.X., 2026. Earth embeddings. URL: <http://arxiv.org/abs/2608.03410>, doi:10.48550/arXiv.2608.03410.
- Yuan, L., Mo, F., Huang, K., Wang, W., Zhai, W., Zhu, X., Li, Y., Xu, J., Nie, J.Y., 2025. OmniGeo: Towards a Multi-modal Large Language Models for Geospatial Artificial Intelligence. URL: <http://arxiv.org/abs/2503.16326>, doi:10.48550/arXiv.2503.16326. arXiv:2503.16326 [cs].
- Zheng, L., Chiang, W.L., Sheng, Y., Zhuang, S., Wu, Z., Zhuang, Y., Lin, Z., Li, Z., Li, D., Xing, E.P., Zhang, H., Gonzalez, J.E., Stoica, I., 2023. Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena. URL: <http://arxiv.org/abs/2306.05685>, doi:10.48550/arXiv.2306.05685. arXiv:2306.05685 [cs].

Table A.5: Shapley

Judge	Metric	Module	Shapley Value	Relative Contribution (%)	Global Delta
H_C	Correctness	R	0.167	16.67%	1.000
		O	0.729	72.92%	
		S	0.104	10.42%	
	Completeness	R	0.396	12.18%	3.250
		O	2.083	64.10%	
		S	0.771	23.72%	
	Relevance	R	0.229	9.65%	2.375
		O	1.854	78.07%	
		S	0.292	12.28%	
	Utility	R	0.271	8.67%	3.125
		O	2.208	70.67%	
		S	0.646	20.67%	
	Grounding	R	0.208	12.82%	1.625
		O	1.208	74.36%	
		S	0.208	12.82%	
H_E	Correctness	R	0.021	3.33%	0.625
		O	0.583	93.33%	
		S	0.021	3.33%	
	Completeness	R	-0.021	-0.98%	2.125
		O	1.541	72.55%	
		S	0.604	28.43%	
	Relevance	R	-0.146	-7.78%	1.875
		O	1.479	78.89%	
		S	0.542	28.89%	
	Utility	R	-0.104	-6.41%	1.625
		O	1.333	82.05%	
		S	0.396	24.36%	
	Grounding	R	0.042	33.33%	0.125
		O	0.042	33.33%	
		S	0.042	33.33%	
LLM_Q	Correctness	R	-0.146	-14.58%	1.000
		O	0.854	85.42%	
		S	0.292	29.17%	
	Completeness	R	-0.062	0.00%	0.000
		O	0.062	0.00%	
		S	0.000	0.00%	
	Relevance	R	-0.250	-50.00%	0.500
		O	0.625	125.00%	
		S	0.125	25.00%	
	Utility	R	-0.062	-6.25%	1.000
		O	0.812	81.25%	
		S	0.250	25.00%	
	Grounding	R	-0.479	-383.33%	0.125
		O	0.458	366.67%	
		S	0.146	116.67%	
LLM_G	Correctness	R	-0.083	-9.52%	0.875
		O	0.667	76.19%	
		S	0.292	33.33%	
	Completeness	R	-0.021	-4.17%	0.500
		O	0.354	70.83%	
		S	0.167	33.33%	
	Relevance	R	-0.083	-22.22%	0.375
		O	0.292	77.78%	
		S	0.167	44.44%	
	Utility	R	0.000	0.00%	0.750
		O	0.500	66.67%	
		S	0.250	33.33%	
	Grounding	R	-0.208	-41.67%	0.500
		O	0.479	95.83%	
		S	0.229	45.83%	

Table B.6: Assessment Metrics Without Context

Metric & Definition	Score	Descriptor	Criteria
1. Correctness Measures the accuracy and truthfulness of the biological, physical, and geospatial information provided in the response.	5	Excellent	All information is factually accurate, scientifically sound, and geospatially precise. No errors or misconceptions.
	4	Good	Most information is accurate with minor inaccuracies that do not affect the overall validity of the response.
	3	Acceptable	Contains some factual errors or imprecise geospatial information, but core claims are generally correct.
	2	Poor	Multiple significant errors in biological, physical, or geospatial claims. Misleading or questionable accuracy.
	1	Unacceptable	Information is largely incorrect, fabricated, or fundamentally flawed. Critical factual errors throughout.
2. Completeness Evaluates the extent to which the response fully addresses all components of the user's query and the specific task at hand.	5	Excellent	Response comprehensively addresses all aspects of the query with thorough coverage. No missing elements.
	4	Good	Covers most query components adequately. Minor aspects may lack detail but overall complete.
	3	Acceptable	Addresses main query components but omits some secondary aspects. Partial coverage of the task.
	2	Poor	Missing several key components of the query. Response feels incomplete or rushed.
	1	Unacceptable	Fails to address the majority of the query. Critical components are missing entirely.
3. Relevance Assesses the signal-to-noise ratio. Focuses on whether the system is concise and direct, penalizing unnecessary or irrelevant information.	5	Excellent	Highly focused and concise. Every statement directly contributes to answering the query. No filler.
	4	Good	Mostly relevant with minimal tangential information. Stays on topic throughout.
	3	Acceptable	Generally relevant but includes some unnecessary details or mild digressions.
	2	Poor	Contains significant irrelevant content. Noticeable amount of filler or off-topic information.
	1	Unacceptable	Largely irrelevant response. Excessive noise, repetition, or content unrelated to the query.
4. Utility Measures the practical value of the response for a user's decision-making process.	5	Excellent	Highly actionable. Provides clear, practical insights that directly support decision-making.
	4	Good	Useful and applicable. Offers practical value with minor limitations in actionability.
	3	Acceptable	Somewhat useful but lacks specific actionable recommendations. General guidance only.
	2	Poor	Limited practical value. Information is too abstract or lacks applicability to real decisions.
	1	Unacceptable	No practical utility. Response does not support any decision-making or action.

Table B.7: Assessment Metrics With Context

Metric & Definition	Score	Descriptor	Criteria
5. Grounding Measures the degree to which assertions are strictly derived from and supported by the context. Evaluates Evidence-to-Claim ratio.	5	Excellent	All claims are explicitly supported by the context. Proper attribution. No unsupported assertions. High specificity.
	4	Good	Most assertions are grounded in context. Minor inferences are reasonable and context-aligned.
	3	Acceptable	Some claims are grounded while others are based on general knowledge. Mixed sourcing.
	2	Poor	Many assertions lack context support. Model frequently makes claims not supported by provided information.
	1	Unacceptable	Response is largely ungrounded. Contradicts context or fabricates information not present in source material.